# Least squares

## 1 True SGD

### 1.1 Setup

$f(x) = \frac{1}{2}\|\Phi x - y\|^2$, $H = \Phi^\top \Phi$
Let $x_*$ be a solution, $Hx_* = \Phi^\top y$

$$\forall t, \quad y_t = \phi_t^\top x_* + \epsilon_t \text{ with}$$

$$\begin{cases} \epsilon_t \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.} \\ \phi_t \sim \mathcal{N}(0, H) \text{ i.i.d.} \\ \epsilon_t \text{ and } \phi_t \text{ independent} \end{cases}$$

Hence $\mathbb{E}[\phi_t y_t] = \mathbb{E}[\phi_t(\phi_t^\top x_* + \epsilon_t)] = Hx_* + \mathbb{E}[\phi_t \epsilon_t] = \Phi^\top y$

Consider a stochastic gradient

$$g_t = \phi_t(\phi_t^\top x_t - y_t) = \phi_t \phi_t^\top (x_t - x_*) - \epsilon_t \phi_t$$

Then the stochastic gradient descent (SGD) is given by

$$\forall t \geq 0, \quad x_{t+1} - x_* = x_t - \eta_t g_t - x_* = (I - \eta_t \phi_t \phi_t^\top)(x_t - x_*) + \eta_t \epsilon_t \phi_t$$

### 1.2 Calculation of the Risk

In this section, we find a closed form of the excess risk $\mathbb{E}[f(x_t) - f_*]$ as a function of $(\eta_t)$.

Let $Q$ s.t. $H = Q\Lambda Q^\top$ with $\Lambda = \mathrm{Diag}(\lambda)$,
$M_t = Q^\top \mathbb{E}[u_t u_t^\top] Q$

Therefore $\mathbb{E}[f(x_t) - f_*] = \frac{1}{2}\mathbb{E}[\langle u_t, Hu_t\rangle] = \frac{1}{2}\mathrm{Tr}(Hu_t u_t^\top) = \frac{1}{2}\langle M_t, \Lambda\rangle = \frac{1}{2}\langle m_t, \lambda\rangle$
With $m_t$ the diagonal of $M_t$.
We will use the following:

**Lemma 1** (*A corollary of Isserlis's theorem*). *If $X \sim \mathcal{N}(0, H)$ and $M \in \mathbb{R}^{n\times n}$ is a constant matrix,*

$$\mathbb{E}[XX^\top M XX^\top] = \mathrm{Tr}(MH)H + HMH + HM^\top H$$

$$\begin{aligned}
M_{t+1} &= Q^\top \mathbb{E}[((I - \eta_t\phi_t\phi_t^\top)u_t + \eta_t\epsilon_t\phi_t)((I - \eta_t\phi_t\phi_t^\top)u_t + \eta_t\epsilon_t\phi_t)^\top]Q \\
&= Q^\top(\mathbb{E}[(I - \eta_t\phi_t\phi_t^\top)u_t u_t^\top(I - \eta_t\phi_t\phi_t^\top)] + \eta_t^2\sigma^2 H)Q \\
&= Q^\top\mathbb{E}[(u_t - \eta_t\phi_t\phi_t^\top u_t)(u_t - \eta_t\phi_t\phi_t^\top u_t)^\top]Q + \eta_t^2\sigma^2\Lambda \qquad (\epsilon_t \text{ and } \phi_t \text{ independent and } \mathbb{E}\epsilon_t = 0) \\
&= M_t - \eta_t\Lambda M_t - \eta_t M_t\Lambda + \eta_t^2 Q^\top\mathbb{E}[\phi_t\phi_t^\top u_t u_t^\top \phi_t\phi_t^\top]Q + \eta_t^2\sigma^2\Lambda \\
&= M_t - \eta_t\Lambda M_t - \eta_t M_t\Lambda + \eta_t^2(2\Lambda M_t\Lambda + \mathrm{Tr}(\Lambda M_t)\Lambda) + \eta_t^2\sigma^2\Lambda \quad (u_t \text{ and } \phi_t \text{ independent and } \phi_t \text{ is gaussian})
\end{aligned}$$

Hence, $m_{t+1} = m_t - 2\eta_t\Lambda m_t + 2\eta_t^2\Lambda^2 m_t + \eta_t^2\lambda\lambda^\top m_t + \eta_t^2\sigma^2\lambda = \left[(I - \eta_t\Lambda)^2 + \eta_t^2\Lambda^2 + \eta_t^2\lambda\lambda^\top\right] m_t + \eta_t^2\sigma^2\lambda$
Let $A_t = (I - \eta_t\Lambda)^2 + \eta_t^2\Lambda^2 + \eta_t^2\lambda\lambda^\top$ and $A_{s:t} = A_t \ldots A_s$

Then $m_t = A_{0:t-1}m_0 + \sum_{k=0}^{t-1} A_{k+1:t-1}\eta_k^2\sigma^2\lambda$

$$\mathbb{E}[f(x_t) - f_*] = \frac{1}{2}\lambda^\top\left(\prod_{k=0}^{t-1}[(I - \eta_k\Lambda)^2 + \eta_k^2\Lambda^2 + \eta_k^2\lambda\lambda^\top]m_0 + \sum_{k=0}^{t-1}\eta_k^2\sigma^2\prod_{s=k+1}^{t-1}[(I - \eta_s\Lambda)^2 + \eta_s^2\Lambda^2 + \eta_s^2\lambda\lambda^\top]\lambda\right)$$

### 1.3 Approximation

Replace $\lambda\lambda^\top$by $\Lambda^2$. $\lambda\lambda^\top$ and $\Lambda$ have the same diagonal, and this change will make the matrices $A_t$ commutative.
Suppose $\lambda_i = Li^{-\alpha}$

$$\mathbb{E}[f(x_t) - f_*] \approx \frac{1}{2}\sum_{i=1}^{d} Li^{-\alpha}\left(m_{0,i}\prod_{j=0}^{t-1}\left[(1-\eta_j Li^{-\alpha})^2 + 2\eta_j^2 L^2 i^{-2\alpha}\right] + \sum_{k=0}^{t-1}\eta_k^2\sigma^2 Li^{-\alpha}\prod_{j=k+1}^{t-1}\left[(1-\eta_j Li^{-\alpha})^2 + 2\eta_j^2 L^2 i^{-2\alpha}\right]\right)$$

#### 1.3.1 If $\eta_k = \eta$ is constant

We want to model the risk as a simple function of $t$.

$$\mathbb{E}[f(x_t) - f_*] \approx \frac{1}{2}\sum_{i=1}^{d} Li^{-\alpha}\left(\left[(1-\eta Li^{-\alpha})^2 + 2\eta^2 L^2 i^{-2\alpha}\right]^t m_{0,i} + \sum_{k=0}^{t-1}\eta^2\sigma^2 Li^{-\alpha}\left[(1-\eta Li^{-\alpha})^2 + 2\eta^2 L^2 i^{-2\alpha}\right]^{t-1-k}\right) \tag{1}$$

$\sum_{k=0}^{t-1}\eta^2\sigma^2 Li^{-\alpha}\left[(1-\eta Li^{-\alpha})^2 + 2\eta^2 L^2 i^{-2\alpha}\right]^{t-1-k}$ is a geometric sum, therefore

$$\sum_{k=0}^{t-1}\eta^2\sigma^2 Li^{-\alpha}\left[(1-\eta Li^{-\alpha})^2 + 2\eta^2 L^2 i^{-2\alpha}\right]^{t-1-k} = \eta^2\sigma^2 Li^{-\alpha}\frac{1-\left[(1-\eta Li^{-\alpha})^2 + 2\eta^2 L^2 i^{-2\alpha}\right]^t}{1-\left[(1-\eta Li^{-\alpha})^2 + 2\eta^2 L^2 i^{-2\alpha}\right]}$$

We can approximate this sum with an integral, supposing $d \to \infty$ and $m_{0,i} \sim \Delta/i^\beta$

$$\mathbb{E}[f(x_t) - f_*] \approx \frac{1}{2}\int_1^\infty Lu^{-\alpha}\left(\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]^t \Delta u^{-\beta} + \eta^2\sigma^2 Lu^{-\alpha}\frac{1-\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]^t}{1-\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]}\right) du$$

• Bias term: $b = \frac{1}{2}\int_1^\infty Lu^{-\alpha}\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]^t \Delta u^{-\beta} du$

$$b = \frac{L\Delta}{2}\int_1^\infty \frac{1}{u^{\alpha+\beta}} e^{t\ln((1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha})} du$$

Moreover,
$\ln((1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}) = \ln(1 - 2\eta Lu^{-\alpha} + 3\eta^2 L^2 u^{-2\alpha}) = -2\eta Lu^{-\alpha} + (\eta Lu^{-\alpha})^2 + O(u^{-3\alpha})$

$$\begin{aligned}
b &\approx \frac{L\Delta}{2}\int_1^\infty \frac{1}{u^{\alpha+\beta}} e^{(-2\eta Lu^{-\alpha})t} du, \qquad \text{Let } z = u^{-\alpha},\ du = -\frac{1}{\alpha} z^{-\frac{\alpha+1}{\alpha}} dz \\
&\approx \frac{L\Delta}{2}\int_0^1 \frac{z^{1+\frac{\beta}{\alpha}-\frac{1+\alpha}{\alpha}}}{\alpha} e^{-2\eta Lzt} dz \\
&\approx \frac{L\Delta}{2\alpha}\int_0^1 z^{\frac{\beta-1}{\alpha}} e^{-2\eta Lzt} dz \\
&\sim_{t\to\infty} \frac{L\Delta}{2\alpha}\frac{\Gamma(\frac{\beta-1}{\alpha}+1)}{(2\eta Lt)^{\frac{\beta-1}{\alpha}+1}} \qquad \text{(Watson's lemma)}
\end{aligned}$$

• Variance term: $V = \int_1^\infty Lu^{-\alpha} \times \eta^2\sigma^2 Lu^{-\alpha}\frac{1-\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]^t}{1-\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]} du$

$$V = \frac{1}{2}\int_1^\infty \frac{L^2\eta^2\sigma^2}{u^{2\alpha}}\frac{1-\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]^t}{1-\left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]} du$$

We make the rough approximations: $\begin{cases} \left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right] \approx 1 - 2\eta Lu^{-\alpha} \\ 1 - \left[(1-\eta Lu^{-\alpha})^2 + 2\eta^2 L^2 u^{-2\alpha}\right]^t \approx 1 \end{cases}$

Then

$$
\begin{aligned}
V &\approx \frac{1}{2}\int_1^\infty \frac{L^2\eta^2\sigma^2}{u^{2\alpha}}\frac{1}{2\eta L u^{-\alpha}}du \\
&\approx \frac{1}{2}\int_1^\infty \frac{L\eta\sigma^2}{2u^{\alpha}}du \\
&\approx \frac{L\eta\sigma^2}{4(\alpha-1)} \qquad \text{if } \alpha > 1
\end{aligned}
$$

Total approximation (Diagonal and Integral) if $\alpha > 1$:

$$
b + V = \frac{L\Delta}{2\alpha}\frac{\Gamma(\frac{\beta-1}{\alpha}+1)}{(2\eta L t)^{\frac{\beta-1}{\alpha}+1}} + \frac{L\eta\sigma^2}{4(\alpha-1)}
$$

### 1.3.2 linear $\eta_t$

We start by considering the risk approximation derived in Equation (1), expressing it as an integral.

$$
\begin{aligned}
b_t &= \frac{1}{2}\int_1^\infty \frac{L\Delta}{u^{\alpha+\beta}}\prod_{j=0}^{t-1}\left[\left(1-\eta_j\frac{L}{u^\alpha}\right)^2 + 2\eta_j\frac{L^2}{u^{2\alpha}}\right]du \\
&= \frac{1}{2}\int_1^\infty \frac{L\Delta}{u^{\alpha+\beta}}\prod_{j=0}^{t-1}\left[\left(1-\eta\frac{(T-j)L}{Tu^\alpha}\right)^2 + 2\left(\eta\frac{T-j}{T}\right)^2\frac{L^2}{u^{2\alpha}}\right]du \\
&= \frac{L\Delta}{2}\int_1^\infty \frac{1}{u^{\alpha+\beta}}\prod_{j=0}^{t-1}\left[1-\frac{2\eta(T-j)L}{Tu^\alpha} + 3\left(\eta\frac{(T-j)L}{Tu^\alpha}\right)^2\right]du \\
&\approx \frac{L\Delta}{2}\int_1^\infty \frac{1}{u^{\alpha+\beta}}e^{\sum_{j=0}^{t-1}\ln\left(1-\frac{2\eta(T-j)L}{Tu^\alpha}+3\left(\eta\frac{(T-j)L}{Tu^\alpha}\right)^2\right)}du \\
&\approx \frac{L\Delta}{2}\int_1^\infty \frac{1}{u^{\alpha+\beta}}e^{\sum_{j=0}^{t-1}\left(-\frac{2\eta(T-j)L}{Tu^\alpha}\right)}du \\
&\approx \frac{L\Delta}{2}\int_1^\infty \frac{1}{u^{\alpha+\beta}}e^{-\frac{2\eta L}{Tu^\alpha}\left(tT-\frac{t(t-1)}{2}\right)}du
\end{aligned}
$$

Let $z = u^{-\alpha}$. $\quad du = -\frac{1}{\alpha}z^{\frac{\alpha+1}{\alpha}}dz$

$$
\begin{aligned}
b_t &\approx \frac{L\Delta}{2\alpha}\int_0^1 z^{\frac{\beta-1}{\alpha}}e^{-2\eta L\left(t-\frac{t(t-1)}{2T}\right)z}dz \\
&\sim_{t\to\infty} \frac{L\Delta}{2\alpha}\frac{\Gamma\left(\frac{\beta-1}{\alpha}+1\right)}{\left(2\eta L\left(t-\frac{t(t-1)}{2T}\right)\right)^{\frac{\beta-1}{\alpha}+1}}
\end{aligned}
$$

We can use Taylor expansions on the logarithm for the variance:

$$
\begin{aligned}
V_t &= \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}\sum_{k=0}^{t-1}\eta_k^2\sigma^2\frac{L}{u^\alpha}\prod_{j=k+1}^{t-1}\left[\left(1-\eta_j\frac{L}{u^\alpha}\right)^2+2\eta_j^2\frac{L^2}{u^{2\alpha}}\right]du\\
&= \frac{L^2\sigma^2}{2}\sum_{k=0}^{t-1}\eta_k^2\int_1^\infty \frac{1}{u^{2\alpha}}\prod_{j=k+1}^{t-1}\left[1-\frac{2\eta_j L}{u^\alpha}+3\eta_j^2\frac{L^2}{u^{2\alpha}}\right]du\\
&= \frac{L^2\sigma^2\eta^2}{2}\sum_{k=0}^{t-1}\left(\frac{T-k}{T}\right)^2\int_1^\infty \frac{1}{u^{2\alpha}}\exp\left(\sum_{j=k+1}^{t-1}\ln\left(1-\tfrac{2\eta(T-j)L}{Tu^\alpha}+3(\tfrac{\eta(T-j)L}{Tu^\alpha})^2\right)\right)du\\
&\approx \frac{L^2\sigma^2\eta^2}{2}\sum_{k=0}^{t-1}\left(\frac{T-k}{T}\right)^2\int_1^\infty \frac{1}{u^{2\alpha}}\exp\left(\sum_{j=k+1}^{t-1}\left(-\frac{2\eta(T-j)L}{Tu^\alpha}\right)\right)du\\
&\approx \frac{L^2\sigma^2\eta^2}{2}\sum_{k=0}^{t-1}\left(\frac{T-k}{T}\right)^2\int_1^\infty \frac{1}{u^{2\alpha}}\exp\left(-\frac{2\eta L}{Tu^\alpha}(t-k-1)\left(T-\frac{t+k}{2}\right)\right)du\\
&\approx \frac{L^2\sigma^2\eta^2}{2\alpha}\sum_{k=0}^{t-1}\left(\frac{T-k}{T}\right)^2\int_0^1 z^{\frac{\alpha-1}{\alpha}}\exp\left(-\frac{2\eta L}{T}(t-k-1)\left(T-\frac{t+k}{2}\right)z\right)dz
\end{aligned}
$$

The Laplace method applied for every term does not work. Let's try something else: with $t=KT,\ K\in[0,1]$, and $T\to\infty$

$$
\begin{aligned}
V_t/\tfrac{L^2\sigma^2\eta^2}{2\alpha} &= \sum_{k=0}^{KT-1}\left(\frac{T-k}{T}\right)^2\int_0^1 z^{\frac{\alpha-1}{\alpha}}\exp\left(-\frac{2\eta L}{T}(KT-k-1)\left(T-\frac{KT+k}{2}\right)z\right)dz \qquad \text{Let } \ell=KT-k\\
&= \sum_{j=1}^{KT}\left(1-K+\frac{\ell}{T}\right)^2\int_0^1 z^{\frac{\alpha-1}{\alpha}}\exp\left(-2\eta L(\ell-1)\left(1-K+\frac{\ell}{2T}\right)z\right)dz\\
&= KT\int_0^1\int_0^1(1-K+K\xi)^2 z^{\frac{\alpha-1}{\alpha}}\exp\left(-2\eta L(KT\xi-1)\left(1-K+\frac{K\xi}{2}\right)z\right)dzd\xi \quad \text{With } \xi=\frac{\ell}{KT}\\
&= KT\int_0^1\int_0^1(1-K+K\xi)^2 z^{\frac{\alpha-1}{\alpha}}\exp\left(-2\eta L(KT\xi-1)\left(1-K+\frac{K\xi}{2}\right)z\right)dzd\xi
\end{aligned}
$$

If $K=1$:

$$
\begin{aligned}
V_t/\tfrac{L^2\sigma^2\eta^2}{2\alpha} &= T\int_0^1\int_0^1 \xi^2 z^{\frac{\alpha-1}{\alpha}}\exp\left(-\eta L(T\xi-1)\xi z\right)dzd\xi\\
&\approx T\int_0^1\int_0^1 \xi^2 z^{\frac{\alpha-1}{\alpha}}\exp\left(-\eta LT\xi^2 z\right)dzd\xi \qquad \zeta=\xi^2,\ d\zeta=2\sqrt{\zeta}d\xi\\
&\approx T\frac{1}{2}\int_0^1\int_0^1 \zeta^{\frac{1}{2}} z^{\frac{\alpha-1}{\alpha}}\exp\left(-\eta LT\zeta z\right)dzd\zeta \qquad \begin{pmatrix}Y\\Z\end{pmatrix}=\begin{pmatrix}\zeta z\\ z\end{pmatrix},\quad |J|=Z\\
&\approx T\frac{1}{2}\int_0^1\int_Y^1 Y^{\frac{1}{2}}Z^{\frac{\alpha-1}{\alpha}-\frac{1}{2}}\exp\left(-\eta LTY\right)\frac{dZdY}{Z}\\
&\approx T\frac{1}{2}\int_0^1 Y^{\frac{1}{2}}\exp\left(-\eta LTY\right)\int_Y^1 Z^{-\frac{1}{\alpha}-\frac{1}{2}}dZdY\\
&\approx \begin{cases}\frac{T}{2}\int_0^1 Y^{\frac{1}{2}}\exp\left(-\eta LTY\right)\frac{2\alpha}{\alpha-2}\left(1-Y^{\frac{\alpha-2}{2\alpha}}\right)dY & \alpha\neq 2\\ \frac{T}{2}\int_0^1 Y^{\frac{1}{2}}\exp\left(-\eta LTY\right)(-\ln Y)dY & \alpha=2\end{cases}
\end{aligned}
$$

- $\alpha\neq 2$

$$
\begin{aligned}
\frac{T}{2}\int_0^1 Y^{\frac{1}{2}}\exp\left(-\eta LTY\right)\frac{2\alpha}{\alpha-2}\left(1-Y^{\frac{\alpha-2}{2\alpha}}\right)dY &= \frac{T\alpha}{\alpha-2}\left[\int_0^1 Y^{\frac{1}{2}}\exp\left(-\eta LTY\right)dY-\int_0^1 Y^{\frac{\alpha-1}{\alpha}}\exp\left(-\eta LTY\right)\right]dY\\
&\approx \frac{T\alpha}{\alpha-2}\left[\frac{\Gamma(\frac{3}{2})}{(\eta LT)^{\frac{3}{2}}}-\frac{\Gamma(1+\frac{\alpha-1}{\alpha})}{(\eta LT)^{1+\frac{\alpha-1}{\alpha}}}\right]
\end{aligned}
$$

• $\alpha = 2$

$$
\begin{aligned}
-\frac{T}{2}\int_0^1 Y^{\frac{1}{2}}\exp\left(-\eta LTY\right)\ln Y dY &= -\frac{T}{2}\int_0^{\eta LT}\left(\frac{y}{\eta LT}\right)^{\frac{1}{2}}\exp\left(-y\right)\ln(\frac{y}{\eta LT})\frac{dy}{\eta LT}\\
&\approx -\frac{T}{2(\eta LT)^{\frac{3}{2}}}\left[\int_0^{\infty} y^{\frac{1}{2}}\exp\left(-y\right)\ln(y)dy - \ln(\eta LT)\int_0^{\infty} y^{\frac{1}{2}}\exp\left(-y\right)dy\right]\\
&\text{knowing that } \Gamma'(x) = \textstyle\int_0^{\infty} t^{x-1}\ln(t)e^{-t}dt,\\
&\approx -\frac{T}{2(\eta LT)^{\frac{3}{2}}}\left[\Gamma'(\frac{3}{2}) - \ln(\eta LT)\Gamma(\frac{3}{2})\right]
\end{aligned}
$$

Therefore

$$
V_t/\frac{L^2\sigma^2\eta^2}{2\alpha} \approx \begin{cases} \dfrac{T\alpha}{\alpha-2}\left[\dfrac{\Gamma(\frac{3}{2})}{(\eta LT)^{\frac{3}{2}}} - \dfrac{\Gamma\left(\frac{2\alpha-1}{\alpha}\right)}{(\eta LT)^{\frac{2\alpha-1}{\alpha}}}\right] & \alpha \neq 2\\ \dfrac{T}{2(\eta LT)^{\frac{3}{2}}}\left(\Gamma\left(\dfrac{3}{2}\right)\ln(\eta LT) - \Gamma'\left(\dfrac{3}{2}\right)\right) & \alpha = 2 \end{cases}
$$

### 1.3.3 WSD schedule

Let $\eta_j = \begin{cases} \eta & j < T_0 \\ \eta\frac{T-j}{T-T_0} & T_0 \leq j \leq T \end{cases}$

It can be considered a function $(\eta_\xi)_{\xi\in\mathbb{R}}$

$$
\begin{aligned}
b_t &= \frac{1}{2}\int_1^{\infty}\frac{L\Delta}{u^{\alpha+\beta}}\prod_{j=0}^{t-1}\left[\left(1-\eta_j\frac{L}{u^{\alpha}}\right)^2 + 2\eta_j\frac{L^2}{u^{2\alpha}}\right]du\\
&= \frac{1}{2}\int_1^{\infty}\frac{L\Delta}{u^{\alpha+\beta}}\left[\left(1-\eta\frac{L}{u^{\alpha}}\right)^2 + 2\eta^2\frac{L^2}{u^{2\alpha}}\right]^{T_0}\prod_{j=T_0}^{t-1}\left[\left(1-\eta\frac{(T-j)L}{(T-T_0)u^{\alpha}}\right)^2 + 2\left(\eta\frac{T-j}{T-T_0}\right)^2\frac{L^2}{u^{2\alpha}}\right]du\\
&= \frac{1}{2}\int_1^{\infty}\frac{L\Delta}{u^{\alpha+\beta}}\exp\left(T_0\ln\left(\left(1-\eta\frac{L}{u^{\alpha}}\right)^2 + 2\eta^2\frac{L^2}{u^{2\alpha}}\right) + \sum_{j=T_0}^{t-1}\ln\left(\left(1-\eta\frac{(T-j)L}{(T-T_0)u^{\alpha}}\right)^2 + 2\left(\eta\frac{T-j}{T-T_0}\right)^2\frac{L^2}{u^{2\alpha}}\right)\right) d\\
&\approx \frac{1}{2}\int_1^{\infty}\frac{L\Delta}{u^{\alpha+\beta}}\exp\left(T_0\left(-2\eta\frac{L}{u^{\alpha}}\right) + \sum_{j=T_0}^{t-1}\left(-2\eta\frac{(T-j)L}{(T-T_0)u^{\alpha}}\right)\right)du\\
&\approx \frac{L\Delta}{2\alpha}\int_0^1 z^{\frac{\beta-1}{\alpha}}\exp\left(-2\eta LT_0 z - 2\eta Lz\sum_{j=T_0}^{t-1}\frac{T-j}{T-T_0}\right)du\\
&\approx \frac{L\Delta}{2\alpha}\int_0^1 z^{\frac{\beta-1}{\alpha}}\exp\left(-2\eta LT_0 z - 2\eta Lz\frac{(T-t+1)(t-T_0)}{T-T_0}\right)du\\
&\approx \frac{L\Delta}{2\alpha}\frac{\Gamma\left(\frac{\beta-1}{\alpha}+1\right)}{\left[2\eta L\left(T_0 + \frac{(T-t+1)(t-T_0)}{T-T_0}\right)\right]^{\frac{\beta-1}{\alpha}+1}}
\end{aligned}
$$

$$
\begin{aligned}
V_t &= \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}\sum_{k=0}^{t-1}\eta_k^2\sigma^2\frac{L}{u^\alpha}\prod_{j=k+1}^{t-1}\left[\left(1-\eta_j\frac{L}{u^\alpha}\right)^2+2\eta_j^2\frac{L^2}{u^{2\alpha}}\right]du \\
&= \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}\sum_{k=1}^{KT}\eta_{k-1}^2\sigma^2\frac{L}{u^\alpha}\prod_{j=k}^{KT-1}\left[\left(1-\eta_j\frac{L}{u^\alpha}\right)^2+2\eta_j^2\frac{L^2}{u^{2\alpha}}\right]d\xi du \\
&= \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}KT\int_0^1\eta_{KT\xi-1}^2\sigma^2\frac{L}{u^\alpha}\prod_{j=KT\xi}^{KT-1}\left[\left(1-\eta_j\frac{L}{u^\alpha}\right)^2+2\eta_j^2\frac{L^2}{u^{2\alpha}}\right]d\xi du \qquad \xi=\frac{k}{KT} \\
&= \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}KT\int_0^1\eta_{KT\xi-1}^2\sigma^2\frac{L}{u^\alpha}\exp\left(\sum_{j=KT\xi}^{KT-1}\ln\left[\left(1-\eta_j\frac{L}{u^\alpha}\right)^2+2\eta_j^2\frac{L^2}{u^{2\alpha}}\right]\right)d\xi du \\
&\approx \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}KT\int_0^1\eta_{KT\xi-1}^2\sigma^2\frac{L}{u^\alpha}\exp\left(\sum_{j=KT\xi}^{KT-1}-2\eta_j\frac{L}{u^\alpha}\right)d\xi du \\
&\approx \frac{1}{2}\int_1^\infty \frac{L}{u^\alpha}KT\left[\int_0^{\frac{T_0+1}{KT}}\eta^2\sigma^2\frac{L}{u^\alpha}\exp\left(\sum_{j=KT\xi}^{T_0}-2\eta\frac{L}{u^\alpha}+\sum_{j=T_0+1}^{KT-1}-2\eta\frac{T-j}{T-T_0}\frac{L}{u^\alpha}\right)d\xi \right. \qquad \text{Suppose } KT>T_0 \\
&\left. +\int_{\frac{T_0+1}{KT}}^1\eta^2(\frac{T-\xi KT}{T-T_0})^2\sigma^2\frac{L}{u^\alpha}\exp\left(\sum_{j=KT\xi}^{KT-1}-2\eta\frac{T-j}{T-T_0}\frac{L}{u^\alpha}\right)d\xi\right]du \\
&= A_t+B_t
\end{aligned}
$$

$$A_t =$$

$$B_t =$$

### 1.4 Slock

Let us see if the approximation $\lambda\lambda^\top \leftarrow \Lambda^2$ is similar to Slock's model (for $\phi$).
To avoid writing $Q$ each time, we suppose $H=\Lambda$. Then $x_t=u_t$.
Let $(e_i)_i$ be the canonical basis of $\mathbb{R}^d$.

Let $\phi_t = s_t\mu_t$ with $\begin{cases} (\mu_t) \text{ i.i.d., } \forall i\geq 1,\ \mathbb{P}(\mu_t=\sqrt{\mathrm{Tr}\Lambda}e_i)=\frac{\lambda_i}{\mathrm{Tr}\Lambda} \\ (s_t) \text{ i.i.d., } \mathbb{P}(s_t=\pm 1)=\frac{1}{2} \\ (s_t),(\mu_t) \text{ and } (\epsilon_t) \text{ are mutually independent}\end{cases}$

We verify that

$$
\begin{aligned}
\bullet\mathbb{E}[\phi_t] &= \mathbb{E}[s_t]\mathbb{E}[\mu_t] \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
\bullet\mathbb{E}[\phi_t\phi_t^\top] &= \sum_{i\geq 1}\frac{\lambda_i}{\mathrm{Tr}\Lambda}\mathrm{Tr}\Lambda e_ie_i^\top \\
&= \sum_{i\geq 1}\lambda_ie_ie_i^\top \\
&= \Lambda
\end{aligned}
$$

$$
\begin{aligned}
M_{t+1} &= \mathbb{E}[((I-\eta_t\phi_t\phi_t^\top)x_t+\eta_t\epsilon_t\phi_t)((I-\eta_t\phi_t\phi_t^\top)x_t+\eta_t\epsilon_t\phi_t)^\top] \\
&= (\mathbb{E}[(I-\eta_t\phi_t\phi_t^\top)x_tx_t^\top(I-\eta_t\phi_t\phi_t^\top)]+\eta_t^2\sigma^2\Lambda) \\
&= \mathbb{E}[(x_t-\eta_t\phi_t\phi_t^\top x_t)(x_t-\eta_t\phi_t\phi_t^\top x_t)^\top]+\eta_t^2\sigma^2\Lambda \qquad (\epsilon_t \text{ and } \phi_t \text{ independent and } \mathbb{E}\epsilon_t=0) \\
&= M_t-\eta_t\Lambda M_t-\eta_tM_t\Lambda+\eta_t^2\mathbb{E}[\phi_t\phi_t^\top x_tx_t^\top\phi_t\phi_t^\top]+\eta_t^2\sigma^2\Lambda
\end{aligned}
$$

Moreover,

$$\begin{aligned}
\mathbb{E}[\phi_t\phi_t^\top x_t x_t^\top \phi_t\phi_t^\top | x_t] &= \sum_{i\geq 1} \frac{\lambda_i}{\mathrm{Tr}\Lambda} \times (\mathrm{Tr}\Lambda)^2 e_i \underbrace{e_i^\top x_t x_t^\top e_i}_{=x_{t,i}^2} e_i^\top \\
&= \mathrm{Tr}\Lambda \sum_{i\geq 1} \lambda_i x_{t,i}^2 e_i e_i^\top \\
&= \mathrm{Tr}\Lambda \begin{pmatrix} \lambda_1 x_{t,1}^2 & 0 & \cdots \\ 0 & \lambda_2 x_{t,2}^2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \\
&= (\mathrm{Tr}\Lambda)\Lambda X_t^2 \qquad \text{where } X_t = \mathrm{Diag}(x_t)
\end{aligned}$$

So

$$\begin{aligned}
\mathbb{E}[\phi_t\phi_t^\top x_t x_t^\top \phi_t\phi_t^\top] &= (\mathrm{Tr}\Lambda)\Lambda\mathbb{E}[X_t^2] \\
&= (\mathrm{Tr}\Lambda)\Lambda\mathbb{E}\begin{pmatrix} x_{t,1}^2 & 0 & \cdots \\ 0 & x_{t,2}^2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \\
&= (\mathrm{Tr}\Lambda)\Lambda\begin{pmatrix} M_{t,11} & 0 & \cdots \\ 0 & M_{t,22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}
\end{aligned}$$

Then

$$\begin{aligned}
M_{t+1} &= M_t - \eta_t\Lambda M_t - \eta_t M_t\Lambda + \eta_t^2(\mathrm{Tr}\Lambda)\Lambda\begin{pmatrix} M_{t,11} & 0 & \cdots \\ 0 & M_{t,22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} + \eta_t^2\sigma^2\Lambda \\
m_{t+1} &= m_t - 2\eta_t\Lambda m_t + \eta_t^2(\mathrm{Tr}\Lambda)\Lambda m_t + \eta_t^2\sigma^2\lambda \\
&= \left[I + (-2\eta_t + \eta_t^2\mathrm{Tr}\Lambda)\Lambda\right] m_t + \eta_t^2\sigma^2\lambda
\end{aligned}$$

(before: diagonal approximation : $\left[I - 2\eta_t\Lambda + 3\eta_t^2\Lambda^2\right] m_t + \eta_t^2\sigma^2\lambda$)

# 2 2-layer neural network

## 2.1 Setup

$$\begin{array}{ccccc}
\mathbb{R}^n & \xrightarrow{W_1} & \mathbb{R}^m & \xrightarrow{W_2} & \mathbb{R}^p \\
x & \longmapsto & x' & \longmapsto & \hat{y}
\end{array}$$

The loss function is quadratic

$$L(\hat{y}, y) = \frac{1}{2}\|\hat{y} - y\|^2$$

For a given $(x, y)$, we want to minimize

$$f_{x,y} : W \mapsto L(W_2W_1x, y) = \frac{1}{2}\|W_2W_1x - y\|^2$$

The theoretical risk is $R(W) = \mathbb{E}_{x,y}[f_{x,y}(W)]$

We suppose $x \sim \mathcal{N}(0, H)$ and $y = W_2^*W_1^*x + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \Sigma)$ independent of $x$.

Therefore the risk is

$$\begin{aligned}
R(W) &= \mathbb{E}[f_{x,y}(W)] \\
&= \mathbb{E}\left[\frac{1}{2}\|W_2W_1x - W_2^*W_1^*x - \epsilon\|^2\right] \\
&= \mathbb{E}\left[\frac{1}{2}\mathrm{Tr}\big[((W_2W_1 - W_2^*W_1^*)x - \epsilon)((W_2W_1 - W_2^*W_1^*)x - \epsilon)^\top\big]\right] \\
&= \mathbb{E}\left[\frac{1}{2}\mathrm{Tr}\big[(W_2W_1 - W_2^*W_1^*)xx^\top(W_2W_1 - W_2^*W_1^*)^\top - 2(W_2W_1 - W_2^*W_1^*)x\epsilon^\top\big] + \epsilon\epsilon^\top\right] \\
&= \frac{1}{2}\mathrm{Tr}\big[(W_2W_1 - W_2^*W_1^*)H(W_2W_1 - W_2^*W_1^*)^\top\big] + \frac{1}{2}\mathrm{Tr}\Sigma
\end{aligned}$$

### 2.1.1 Gradients

For a given $(x, y)$,

$$\begin{aligned}
Df_{x,y}(W)(\tilde{W}) &= (D[L(\cdot, y)](\hat{y})) \cdot (\tilde{W}_2W_1x + W_2\tilde{W}_1x) \\
&= \langle W_2W_1x - y, W_2\tilde{W}_1x\rangle + \langle W_2W_1x - y, \tilde{W}_2W_1x\rangle \\
&= \langle W_2^\top(W_2W_1x - y)x^\top, \tilde{W}_1\rangle + \langle (W_2W_1x - y)x^\top W_1^\top, \tilde{W}_2\rangle
\end{aligned}$$

So $\begin{cases} \nabla_{W_1} f_{x,y}(W) = W_2^\top(W_2W_1x - y)x^\top \\ \nabla_{W_2} f_{x,y}(W) = (W_2W_1x - y)x^\top W_1^\top \end{cases}$

The SGD at step $t$ is:

$$\begin{aligned}
W_{1,t+1} &= W_{1,t} - \eta_t W_2^\top(W_2W_1x_t - y)x_t^\top \\
&= W_{1,t} - \eta_t W_{2,t}^\top((W_{2,t}W_{1,t} - W_2^*W_1^*)x_t - \epsilon_t)x_t^\top \\
&= W_{1,t} - \eta_t W_{2,t}^\top(W_{2,t}W_{1,t} - W_2^*W_1^*)x_tx_t^\top + \eta_t W_{2,t}^\top\epsilon_t x_t^\top
\end{aligned}$$

$$\begin{aligned}
W_{2,t+1} &= W_{2,t} - \eta_t(W_{2,t}W_{1,t}x_t - y)x_t^\top W_{1,t}^\top \\
&= W_{2,t} - \eta_t((W_{2,t}W_{1,t} - W_2^*W_1^*)x_t - \epsilon_t)x_t^\top W_{1,t}^\top \\
&= W_{2,t} - \eta_t(W_{2,t}W_{1,t} - W_2^*W_1^*)x_tx_t^\top W_{1,t}^\top + \eta_t\epsilon_t x_t^\top W_{1,t}^\top
\end{aligned}$$

Let $w_t = W_{2,t}W_{1,t} - W_2^*W_1^*$ and $\delta_t = \hat{y}_t - y_t = w_tx_t - \epsilon_t$

$$\begin{aligned}
w_{t+1} &= (W_{2,t} - \eta_t\delta_t x_t^\top W_{1,t}^\top)(W_{1,t} - \eta_t W_{2,t}^\top\delta_t x_t^\top) - W_2^*W_1^* \\
&= w_t - \eta_t(\delta_t x_t^\top W_{1,t}^\top W_{1,t} + W_{2,t}W_{2,t}^\top\delta_t x_t^\top) + \eta_t^2\delta_t x_t^\top W_{1,t}^\top W_{2,t}^\top\delta_t x_t^\top
\end{aligned}$$

$$\begin{aligned}
R(W_t) &= \frac{1}{2}\mathrm{Tr}\big[w_tHw_t^\top\big] + \frac{1}{2}\mathrm{Tr}\Sigma \\
&= \frac{1}{2}\mathrm{Tr}\big[Hw_t^\top w_t\big] + \frac{1}{2}\mathrm{Tr}\Sigma
\end{aligned}$$

Let $M_t = \mathbb{E}[w_t^\top w_t]$. Then $\mathbb{E}[R(W_t) - \frac{1}{2}\mathrm{Tr}\Sigma] = \frac{1}{2}\langle H, M_t\rangle$
Let us compute $M_{t+1}$

$$\begin{aligned}
M_{t+1} = \mathbb{E}[&\big(w_t - \eta_t(\delta_t x_t^\top W_{1,t}^\top W_{1,t} + W_{2,t}W_{2,t}^\top\delta_t x_t^\top) + \eta_t^2\delta_t x_t^\top W_{1,t}^\top W_{2,t}^\top\delta_t x_t^\top\big)^\top \\
&\times \big(w_t - \eta_t(\delta_t x_t^\top W_{1,t}^\top W_{1,t} + W_{2,t}W_{2,t}^\top\delta_t x_t^\top) + \eta_t^2\delta_t x_t^\top W_{1,t}^\top W_{2,t}^\top\delta_t x_t^\top\big)]
\end{aligned}$$

We may calculate the expectation for each term.
We know that $\mathbb{E}[\delta_t x_t^\top | w_t] = \mathbb{E}[w_tx_tx_t^\top | w_t] - \mathbb{E}[\epsilon_t x_t^\top | w_t] = w_tH$

$$
\begin{aligned}
&\mathbb{E}[W_{1,t}^\top W_{1,t} x_t \delta_t^\top w_t] = \mathbb{E}[W_{1,t}^\top W_{1,t} H w_t^\top w_t] \\
&\mathbb{E}[x_t \delta_t^\top W_{2,t}^\top W_{2,t} w_t] = \mathbb{E}[H w_t^\top W_{1,t}^\top W_{1,t} w_t] \\
&\mathbb{E}[W_{1,t}^\top W_{1,t} x_t \delta_t^\top \delta_t x_t^\top W_{1,t}^\top W_{1,t}] = ... \\
&\mathbb{E}[x_t \delta_t^\top W_{2,t} W_{2,t}^\top W_{2,t} W_{2,t}^\top \delta_t x_t^\top] = ...
\end{aligned}
$$